In [ ]:
```
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#Roll No : A33
#Batch : A - 2
#Assignment No : 09 [EDA Lab -> Data Balancing]
```

In [2]:
```
pip install -U imbalanced-learn
```

```
Requirement already satisfied: imbalanced-learn in c:\users\student\anaco
nda3\lib\site-packages (0.14.2)
Requirement already satisfied: numpy<3,>=1.25.2 in c:\users\student\anaco
nda3\lib\site-packages (from imbalanced-learn) (1.26.4)
Requirement already satisfied: scipy<2,>=1.11.4 in c:\users\student\anaco
nda3\lib\site-packages (from imbalanced-learn) (1.11.4)
Requirement already satisfied: scikit-learn<2,>=1.4.2 in c:\users\studen
t\anaconda3\lib\site-packages (from imbalanced-learn) (1.9.0)
Requirement already satisfied: sklearn-compat<0.2,>=0.1.6 in c:\users\stu
dent\anaconda3\lib\site-packages (from imbalanced-learn) (0.1.6)
Requirement already satisfied: joblib<2,>=1.2.0 in c:\users\student\anaco
nda3\lib\site-packages (from imbalanced-learn) (1.6.0)
Requirement already satisfied: threadpoolctl<4,>=2.0.0 in c:\users\studen
t\anaconda3\lib\site-packages (from imbalanced-learn) (3.6.0)
Requirement already satisfied: cloudpickle>=3.0 in c:\users\student\anaco
nda3\lib\site-packages (from joblib<2,>=1.2.0->imbalanced-learn) (3.1.2)
Requirement already satisfied: narwhals>=2.0.1 in c:\users\student\anacon
da3\lib\site-packages (from scikit-learn<2,>=1.4.2->imbalanced-learn)
(2.25.0)
Note: you may need to restart the kernel to use updated packages.
```

In [3]:
```
import pandas as pd
import numpy as np
from imblearn.under_sampling import RandomUnderSampler
from collections import Counter
```

In [32…
```
df = pd.DataFrame({
    'Maths': [85, 78, 45, 90, 88, 76, 35, 92, 81, 79],
    'Physics': [80, 75, 40, 85, 82, 70, 30, 95, 84, 73],
    'Chemistry': [78, 82, 38, 88, 85, 74, 32, 91, 79, 75],
    'Result': [1, 1, 0, 1, 1, 1, 0, 1, 1, 1]
})
df
```

Out[32]:

	Maths	Physics	Chemistry	Result
0	85	80	78	1
1	78	75	82	1
2	45	40	38	0
3	90	85	88	1
4	88	82	85	1
5	76	70	74	1
6	35	30	32	0
7	92	95	91	1
8	81	84	79	1
9	79	73	75	1

In [33…
```
# Q.1 Implement UnderSampling technique for handling imbalanced dataset.
x = df[['Maths','Physics','Chemistry']]
y = df['Result']

print("Before Under-Sampling:",Counter(y))
```

```
Before Under-Sampling: Counter({1: 8, 0: 2})
```

In [34…
```
rus = RandomUnderSampler(random_state = 42)
x_under,y_under = rus.fit_resample(x,y)
```

In [35…
```
print("After Under-Sampling:",Counter(y_under))
```

```
After Under-Sampling: Counter({0: 2, 1: 2})
```

In [36…
```
# Q.2 Implement OverSampling technique for handling imbalanced dataset.
from imblearn.over_sampling import RandomOverSampler
X = df[['Maths','Physics','Chemistry']]
Y = df['Result']

print("Before Over-Sampling:",Y.value_counts())
```

```
Before Over-Sampling: Result
1    8
0    2
Name: count, dtype: int64
```

In [37…
```
ros = RandomOverSampler(random_state = 42)
X_over,Y_over = ros.fit_resample(X,Y)
```

In [38…
```
print("After Over-Sampling:",Y_over.value_counts())
```

```
After Over-Sampling: Result
1    8
0    8
Name: count, dtype: int64
```